(a) Show that in 3D the Green's function for a half-space $D = \{z > 0\}$ is

$$G(\mathbf{x}, \mathbf{x}_0) = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}_0|} + \frac{1}{4\pi|\mathbf{x} - \mathbf{x}^*{}_0|}$$

where $\mathbf{x} = (x, y, z)$ and $\mathbf{x}^* = (x, y, -z)$.

Let's verify it by checking each of the three properties of G:

Property 1. $G(\vec{x})$ possesses continuous second derivatives and $\Delta G = 0$ in D, except at the point $\vec{x} = \vec{x}_0$.

Clearly, G is finite and differentiable except at $\vec{x}_0$.

Also, $\Delta G = 0$.

Property 2. $G(\vec{x}) = 0$ for $\vec{x} \in \partial D$.

Let $\vec{x} \in \partial D$, so $z = 0$. Hence $|\vec{x} - \vec{x}_0| = |\vec{x} - \vec{x}_0^*| = \sqrt{(x-x_0)^2 + (y-y_0)^2 + z_0^2}$.

Thus $G(\vec{x}, \vec{x}_0) = 0$

Property 3. The function $G(\vec{x}) + \frac{1}{4\pi|\vec{x} - \vec{x}_0|}$ is finite at $\vec{x}_0$ and has continuous second derivatives everywhere and is harmonic at $\vec{x}_0$.

Because $\vec{x}_0^*$ is outside the half-space D, the function $\frac{1}{4\pi|\vec{x} - \vec{x}_0^*|}$ has no singularity inside the domain, so that G has the proper singularity at $\vec{x}_0$.

(b) Show that the solution of Laplace equation

$$\begin{cases} \triangle u = 0, & \text{in } D \\ u = h, & \text{on } \partial D \end{cases} \tag{1}$$

is represented by

$$u(\mathbf{x}_0) = \iint_{\partial D} \frac{1}{2\pi} \frac{z_0}{|\mathbf{x} - \mathbf{x}^*{}_0|^3} h(\mathbf{x}) dS$$

[Hint]

$$u(\mathbf{x}_0) = \iint_{\partial D} \frac{\partial G(\mathbf{x}, \mathbf{x}_0)}{\partial n} h(\mathbf{x}) dS$$

solves Eq. (1).

From hint, we need to prove $\frac{\partial G(\vec{x}, \vec{x}_0)}{\partial n} = \frac{1}{2\pi} \frac{z_0}{|\vec{x} - \vec{x}_0^*|^3}$.

In half-space $D = \{z > 0\}$, $n = (0, 0, -1)$

So, along ∂D, $\frac{\partial G}{\partial n} = \nabla G \cdot n = -\frac{\partial G}{\partial z}\Big|_{z=0}$

$$= \frac{1}{4\pi} \left(\frac{z + z_0}{|\vec{x} - \vec{x}_0^*|^3} - \frac{z - z_0}{|\vec{x} - \vec{x}_0^*|^3} \right) = \frac{1}{2\pi} \frac{z_0}{|\vec{x} - \vec{x}_0^*|^3}$$

(c) For two dimensional case, show that the Green's function for the half-plane $D = \{y > 0\}$ is

$$G(\mathbf{x}, \mathbf{x}_0) = \frac{1}{2\pi} \log |\mathbf{x} - \mathbf{x}_0| - \frac{1}{2\pi} \log |\mathbf{x} - \mathbf{x}^*{}_0|$$

where $\mathbf{x} = (x, y)$ and $\mathbf{x}^* = (x, -y)$. Represent the solution of Eq. (1).

First, let's check each of the three properties of G:

Property 1. $G(\vec{x})$ possesses continuous second derivatives and $\Delta G = 0$ in D, except at the point $\vec{x} = \vec{x}_0$.

Clearly, G is finite and differentiable except at $\vec{x}_0$.

Also, $\Delta G = 0$.

Property 2. $G(\vec{x}) = 0$ for $\vec{x} \in \partial D$.

Let $\vec{x} \in \partial D$, so $y = 0$. Hence $|\vec{x} - \vec{x}_0| = |\vec{x} - \vec{x}_0^*| = \sqrt{(x - x_0)^2 + y_0^2}$

Thus $G(\vec{x}, \vec{x}_0) = 0$.

Property 3. The function $G(\vec{x}) - \frac{1}{2\pi} \log |\vec{x} - \vec{x}_0|$ is finite at $\vec{x}_0$ and has continuous second derivatives everywhere and is harmonic at $\vec{x}_0$.

Because $\vec{x}_0^*$ is outside the half-space D, the function $-\frac{1}{2\pi} \log |\vec{x} - \vec{x}_0^*|$ has no singularity inside the domain, so that G has the proper singularity at $\vec{x}_0$.

To present the solution of Eq.(1), we need to consider a large semicircle: $\iint (u\nabla^2 G - G\nabla^2 u)\,dA = \oint (u\nabla G - G\nabla u)\cdot \hat{n}\,ds = \int_{-\infty}^{\infty} \left(G\frac{\partial u}{\partial y} - u\frac{\partial G}{\partial y}\right)\Big|_{y=0} dx$

Since for the wall the outward unit normal is $\hat{n} = -\hat{j}$, and since the contribution at ∞ tends to vanish if $u \to 0$ sufficiently fast (in particular, $\lim_{r\to\infty}(u - r\ln r \frac{\partial u}{\partial r}) = 0$)

Substituting the defining differential equations and interchanging $\vec{x}$ and $\vec{x}_0$ (using the symmetry of $G(\vec{x}, \vec{x}_0)$) shows that

$$u(\vec{x}) = \iint f(\vec{x}_0)\, G(\vec{x}, \vec{x}_0)\, dA_0 - \int_{-\infty}^{\infty} h(x_0) \frac{\partial}{\partial y_0} G(\vec{x}, \vec{x}_0)\Big|_{y_0=0} dx_0.$$

Since $G = 0$ on $y = 0$.

$$\frac{\partial G(\vec{x}, \vec{x}_0)}{\partial y_0} = \frac{1}{4\pi}\left[\frac{-2(y-y_0)}{(x-x_0)^2+(y-y_0)^2} - \frac{2(y+y_0)}{(x-x_0)^2+(y+y_0)^2}\right].$$

Evaluating this at $y_0 = 0$ yields

$$\frac{\partial G(\vec{x}, \vec{x}_0)}{\partial y_0}\Big|_{y_0=0} = -\frac{y/\pi}{(x-x_0)^2+y^2}$$

So, $u(\vec{x}) = \int_{-\infty}^{\infty} h(x_0) \frac{y/\pi}{(x-x_0)^2+y^2}\, dx_0$.